

Styled Algorithmic Trading and the MV-MVP Style

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Abstract

Except for pre-trade assessment, algorithmic trading for optimal execution has to be *dynamically* adapted to real-time market environments and inventory positions. This makes *dynamic programming* (DP) the most natural approach. Due to the curse of dimensionality, however, it is highly challenging to design a rigorous DP strategy that invites a tractable solution procedure. *Approximate dynamic programming* (ADP) has hence emerged as a practical solution, and the proposed framework of styled trading follows the ADP paradigm. A styled model makes a particular choice for its state variables as well as the parametric forms of its trading policies. Its calibration then becomes a nonlinear, constrained, but lower-dimensional optimization problem. The current framework also goes beyond the *additive* objectives normally required by a traditional DP strategy and its associated Bellman solution procedure. It works with the mean-variance utilities of Almgren and Chriss (*J. Risk*, **3**, 2000) to effectively express risk aversion of a typical trader. The new framework is demonstrated through building one particular style called MV-MVP, i.e., the *mean-variance* (MV) objective formulated upon the state variables of *moneyness* and *volume participation* (MVP). The MV-MVP style generalizes the VWAP strategy by facilitating dynamic reactions to moneyness and by embodying the popular street practice of trading aggressively or passively while in the money. Numerical calibration based on *stochastic gradient descent* and *binomial-tree gradient descent* is discussed. Simulated dynamic trading paths help illustrate the general behavior of the MV-MVP style.

Keywords: Dynamic programming, style, moneyness, aggressive, passive, participation, parametric, stochastic, gradient descent, binomial trees.

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1. Introduction

For major institutional clients such as mutual funds, retirement funds, or hedge funds, algorithmic trading for optimal execution helps liquidate or acquire big positions with limited impacts on both the markets and the clients. The execution houses (e.g., broker-dealers and agencies) in return generate commission revenues to cover their costs on IT infrastructures (e.g., servers or data warehouses), as well as on model and analytics development. Therefore, execution motivated algorithmic trading has become a crucial financial service for modern electronic markets.

The most striking characteristics of algorithmic trading, as compared with the one-shot trading activities of small individual investors, is its constant flow of trading decisions made throughout the entire execution horizon. The key is to decide how to best “slice” the entire big order into smaller “child” orders to be progressively executed. Ideally such decisions have to resonate with the real-time dynamics of the markets and inventory positions. Dynamic programming (DP) naturally becomes the central approach to optimal algorithmic trading. This could be witnessed from the pioneering works of Bertsimas and Lo [6], and Huberman and Stanzl [9], to more recent ones by Almgren [1], Bouchard, Dang, and Lehalle [7], and Azencott et al. [4], just to name a few. Static algorithms, e.g., those by Almgren and Chriss [2], Huberman and Stanzl [9], Obizhaeva and Wang [12], Avellaneda and Stoikov [3], Kissell and Malamut [11], Gatheral [8], or Shen [17], can provide useful pre-trade estimation for clients. They nevertheless do not respond to fresh market information and thus cannot be designated as actual execution strategies.

While it provides a natural paradigm, DP does impose a number of challenges on both modeling and computing. For instance, since the *Flash Crash* in the US markets on May 6th, 2011, it has been clear that trading risks can be significantly curbed via setting up even simple constraints like price or volume limits. Thus fundamentally algorithmic trading has to be a *constrained* DP problem, and closed-form solutions generally do not exist. In Section 2, we will discuss in more technical details several outstanding challenges in implementing a self-contained DP trading strategy, including, e.g., (a) defining proper DP objectives meaningful to general traders, (b) properly handling key trading constraints, and (c) efficiently dealing with the explosion of dimensions.

There are two main approaches to answering these challenges *in practice*, and both are in the approximation sense. The first is to *periodically* call a *static strategy*, e.g., every 20 minutes or whenever certain events trigger. This is a heuristic way to emulate dynamic trading via static strategies like that of Almgren and Chriss [2]. Traditionally many execution houses have adopted this approach as it is much easier to integrate algorithmic trading with efficient static commercial optimizers such as IBM CPLEX or MOSEK. This heuristic approach can take good care of client constraints but generally lacks a coherent theoretical foundation.

The second way turns to the method of *approximate dynamic programming* (ADP). ADP is currently an active and rapidly evolving research area (e.g., [5,

15]). ADP aims at breaking the *curse of dimensionality* via properly designed approximation schemes. The approximations may be made for the objectives, state spaces, control actions, state transitions, the policies, or the Bellman equations. Despite such a vast degree of freedom, there exist no ADP design methods that work universally well for arbitrarily given DP problems. The design of an effective ADP scheme often relies on the particular characteristics of the tasks at hand as well as the insights of users (e.g., [18]).

The current work belongs to ADP in the above general sense. We define a style to be a particular choice of the following two components: (1) state variables, and (2) policies built upon these variables. Section 3 shall give a more technical account. Traditionally DP trading can only work with *additive* objectives. A style defined this ways allows a much broader class of trading objectives, including, e.g., the mean-variance utilities of Almgren and Chriss [2].

The remaining sections demonstrate styled trading via the detailed construction of a particular style using the *mean-variance* objective and the two state variables of *moneyiness and volume participation*. For convenience, this particular style is called the MV-MVP strategy. In Section 4 we define the two dimensionless state variables - *moneyiness* and *forward PoV*, and also derive their state transition equations. In Section 5 we introduce the mean-variance objective for the *augmented* implementation shortfall (IS) [13], and define a particular parametric form of trading policies. The popular VWAP strategy can be considered as a special scenario of MV-MVP when all the four parameters are set to zero. Model calibration is discussed in Section 6, in which a generic instance of the MV-MVP style is also worked out.

Below are some important settings or assumptions for the current work.

- (A) The trading horizon $(0, T]$ is assumed to be an intraday time window, e.g., from 11:15 am to 1:30 pm, within the *continuous session*. The starting time is always denoted by $t = 0$. The current work does not cover any auction sessions such as the opens or closes.
- (B) As in most aforementioned works in the literature, we are primarily concerned with optimal decisions on setting the trade sizes for arrival bins. Let Δt denote an internally chosen child window size, e.g., 5 minutes. The trading horizon $(0, T]$ is then “sliced” into N child bins:

$$(t_0, t_1, \dots, t_N) = (0, \Delta t, 2\Delta t, \dots, N\Delta t = T).$$

At each time t_n , the decision is to be made for the number of shares q_n to trade over the arrival bin $(t_n, t_{n+1}]$.

- (C) The current work does not address how to actually execute q_n shares within an arrival bin $(t_n, t_{n+1}]$. This is called the process of *Child Order Placement* (COP). A COP strategy often involves the following key components: (a) further slicing each child order into grandchild orders, (b) timing and anti-gaming of placing each grandchild order, (c) selecting market or limit order types for each grandchild order, and (d) for limit orders determining book sitting levels, waiting times, and cancel-and-replace policies. In particular, one observes that the bin size Δt should be big enough in order to

accommodate any COP limit-order strategies. Otherwise, time may be too short for the limit orders to be traded through.

2. Challenges in Dynamic Trading

We first discuss some major challenges that a traditional DP strategy has to face, in order to motivate styled DP trading.

For each trading bin $(t_n, t_{n+1}]$, let $\mathbf{x}_n = (x_{n,1}, \dots, x_{n,d})$ denote the market or inventory state variables observed at t_n , and q_n the number of shares to be traded (i.e., the control variable). For recursive DP trading to work, the path-dependent total cost has to take the additive form:

$$J_0(q_0, \dots, q_{N-1}; \boldsymbol{\omega} \mid \mathbf{x}_0) = \sum_{n=0}^{N-1} \beta^n g_n(\mathbf{x}_n, q_n) + \beta^N g_N(\mathbf{x}_N), \quad (1)$$

where $\boldsymbol{\omega}$ denotes a market path, and $\beta \geq 0$ a potential discount factor. For a single-factor market, one has $\boldsymbol{\omega} = (\omega_0, \dots, \omega_{N-1})$ with ω_n denoting the scalar random market movement over $(t_n, t_{n+1}]$. The stagewise cost $g_n(\cdot, \cdot)$ and terminal cost $g_N(\cdot)$ may be stage dependent, as indicated by the subscripts. Let $\mathcal{D}_n(\mathbf{x}_n)$ denote the permissible action space for trading bin n , so that one demands $q_n \in \mathcal{D}_n(\mathbf{x}_n)$. Dynamical trading is to seek a series of optimal policies

$$\varphi_0^*(\cdot), \varphi_1^*(\cdot), \dots, \varphi_{N-1}^*(\cdot),$$

such that

$$(\varphi_0^*, \dots, \varphi_{N-1}^*) = \arg \min_{(\varphi_0, \dots, \varphi_{N-1})} E_{\boldsymbol{\omega}}[J_0(q_n = \varphi_n(\mathbf{x}_n), 0 \leq n < N; \boldsymbol{\omega} \mid \mathbf{x}_0)]. \quad (2)$$

At the beginning time t_n of $(t_n, t_{n+1}]$, define the *spot* cost

$$J_n(q_n, \dots, q_{N-1}, \boldsymbol{\omega} \mid \mathbf{x}_n) = \sum_{k=n}^{N-1} \beta^{k-n} g_k(\mathbf{x}_k, q_k) + \beta^{N-n} g_N(\mathbf{x}_N),$$

and the *spot* value function:

$$v_n(\mathbf{x}_n) = \min_{(\varphi_n, \dots, \varphi_{N-1})} E_{\boldsymbol{\omega}}[J_n(q_k = \varphi_k(\mathbf{x}_k), n \leq k < N; \boldsymbol{\omega} \mid \mathbf{x}_n)]. \quad (3)$$

Then one has the backward recursive Bellman equation: for $n = N-1, \dots, 0$,

$$v_n(\mathbf{x}_n) = \min_{q_n} \{g_n(\mathbf{x}_n, q_n) + \beta E_{\omega_n}[v_{n+1}(F_n(\mathbf{x}_n, q_n, \omega_n))]\}. \quad (4)$$

Here $\mathbf{x}_{n+1} = F_n(\mathbf{x}_n, q_n, \omega_n)$ is the state transition equation with stage specific market noise ω_n . The main challenges of dynamic algorithmic trading are summarized as follows.

- (a) **Pros and Cons of Additive Costs.** Additivity has induced the whole Bellman machinery (i.e., Eqn. (4)), but also severely restricted the available class of cost functions. It forces one to either work with the risk-neutral setting or adopt additive risk measures such as the *total quadratic variance*. Both do not explain well to traders who typically request the lowest average implementation shortfalls (IS) given a tolerable risk level. For them, the mean-variance objectives of Almgren and Chriss [2] are more pertinent even though they are non-additive.
- (b) **Completion.** A typical trader prefers an algorithm to complete the entire order within a specified trading horizon, say, from 10:00 am to 11:30 am. Most existing works achieve it by setting q_{N-1} of the last bin ($t_{N-1}, t_N = T$) to be the entire remaining order [6]. In reality, this may well deplete the entire market over the last bin if the outstanding order significantly dwarfs the market. One simple mechanism in favor of completion is to enforce a minimum participation rule:

$$q_n = \varphi_n(\mathbf{x}_n) \geq \text{minPoV} \cdot V_n \quad (5)$$

where V_n denotes the (projected) market volume for trading bin $(t_n, t_{n+1}]$, and minPoV some minimum rate of *participation of volume*, e.g., 5%. Based on the size of the initial order and the trading horizon, such a rule can at least guarantee certain portion of the order to be completed, if not the all. Constraints like this have been hardly incorporated due to the generic difficulty in handling constraints in conventional DP trading.

- (c) **Explosion of Dimensions.** Except for simple and often constraint-free models, e.g., LQR [5], there are typically no closed forms for either the value functions $v_n(\cdot)$'s or the optimal policies $\varphi_n(\cdot)$'s. Let $\mathcal{A}_n(\mathbf{x}_n) = \mathcal{A}_n \subseteq R^d$ denote the state space for $\mathbf{x}_n \in R^d$. Each optimal policy φ_n^* is a mapping:

$$\varphi_n^* : \mathcal{A}_n(\mathbf{x}_n) \rightarrow \mathcal{D}_n(\mathbf{x}_n), \quad \mathbf{x}_n \rightarrow q_n^* = \varphi_n^*(\mathbf{x}_n).$$

Even in the discretized setting, the search space for any single policy φ_n^* would have size L^M , assuming the state space $\mathcal{A}_n$ is discretized to M states, and the action space $\mathcal{D}_n$ to L actions. The search space is usually humongous.

- (d) **Constraints Handling.** Besides the completion constraint, typical traders also impose the following two constraints:
- (4.A) **Monotonicity.** That is, for a buy order, one demands: $q_n \geq 0$ for any n . Similarly, for a sell order, one requires: $q_n \leq 0$ for each n .
- (4.B) **Volume Participation Limit.** Most traders prefer to cap the maximum rate of PoV, such that:

$$q_n \leq \text{maxPoV} \cdot V_n, \quad 0 \leq n < N. \quad (6)$$

This is typically used as the ultimate gate keeper for avoiding market crash, and is especially important for illiquid securities.

Such constraints have not been rigorously enforced in most traditional DP models due to the universal difficulty in accommodating constraints in DP.

3. Styled Dynamic Algorithm Trading

As the terminology “style” has not yet been commonly circulated in the industry, we first give a more detailed account about styles and style designing.

We define a “style” to be a user’s particular choice of the two key factors:

- (a) State variables $\mathbf{x}_n = (x_{n,1}, \dots, x_{n,d})$ for each bin $(t_n, t_{n+1}]$, based on which trading decisions are solely made.
- (b) A particular parametric form of trading policies:

$$q_n = \varphi_n(\mathbf{x}_n \mid \Theta), \quad n = 1, \dots, N-1, \quad (7)$$

based on the state variables $\mathbf{x}_n$, and a finite set of parameters Θ . The set of parameters can be further split to two sets: $\Theta = \Theta_i \cup \Theta_e$. Θ_e denotes the set of trading control parameters exposed *externally* to traders. For example, most clients prefer to cap the volume participation level, e.g. $\text{maxPoV} = 25\%$. Θ_i denotes the subset of remaining parameters that add further trading control but are *internally* optimized.

As an example, in the execution industry one often hears the terms “aggressive in the money (ITM)” or “passive in the money”. Here,

- (i) Moneyiness must be one of the state variables, which we shall elaborate on later. For the moment, let us denote moneyiness by δ_n .
- (ii) Suppose the buying strategy can be expressed by $q_n = \varphi_n(\delta_n, \dots \mid \Theta)$, and suppose that for a buy order, ITM is defined as $\delta_n < 0$. Then the style of “being aggressive ITM” means that φ_n is monotonically decreasing on δ_n so that one trades aggressively when δ_n becomes more negative.

Since styled trading works with a particular parametric class of policies, it can only be suboptimal compared with the full engine of DP trading. But the gains are also substantial.

- (A) Significant complexity reduction. Under styled algorithmic trading, one is only required to calibrate a few or several internal parameters Θ_i , via a lower dimensional optimization problem.
- (B) Flexible handling of constraints. For example, most traders prefer to set the volume limit maxPoV . It suffices to choose a style in the form of: (with $a \wedge b = \min(a, b)$)

$$q_n = \varphi(\mathbf{x}_n \mid \Theta) = g_n(\mathbf{x}_n \mid \Theta) \wedge (\text{maxPoV} \cdot V_n),$$

where g is a “free” style, and V_n the projected market volume.

To illustrate the main ideas of styled algorithmic trading, for the rest of the paper, we shall focus on designing one specific style called the MV-MVP model. It is built upon the *mean-variance* (MV) objective and the state variables of *moneyiness* and *volume participation* (MVP). To some degree, the MV-MVP style extends the popular VWAP algorithm by facilitating extra dynamic response to moneyiness.

For simplicity, hereafter we shall use Θ to denote only the *internal* model parameters Θ_i . External parameters Θ_e are trading parameters set by traders and need no computation. Model calibration then means to calibrate Θ . For MV-MVP, Θ only contains four scalar parameters (a, b, c, d) .

4. States and Dynamics Based on Moneyness and Forward PoV

Let $(0, T]$ denote the trading horizon for buying or selling Q shares in total. Since the current model assumes no biases between buy and sell, we shall fix the running example to be a buy order. Following the Introduction, a DP algorithm decides the number of shares q_n to trade over each arrival bin $(t_n, t_{n+1} = t_n + \Delta t]$, with Δt being 1 minute or 5 minutes, for instance. It then relies upon a sound *Child Order Placement* (COP) strategy for actually executing q_n shares over the bin. A typical COP strategy involves a complex flow of actions on further grandchild order slicing, limiting order placement, and aggressive spread crossing, etc, as addressed in the Introduction.

4.1. State Variables: Moneyness and Forward PoV

Let p_n denote the opening price at t_n for each bin $(t_n, t_{n+1}]$. In reality, it could be the mid price of the National Best Bid and Offer (NBBO) at t_n , for example. For $n = 0$, p_0 is the initial arrival price for the entire execution. At each t_n , the state variable “moneyness” is defined by:

$$\delta_n = \frac{p_n - p_0}{p_0} = \frac{p_n}{p_0} - 1.0, \quad n = 0, 1, \dots, N-1. \quad (8)$$

Similarly, for analytical or computational purposes, we define the *posterior* moneyness $\bar{\delta}_n$ by:

$$\bar{\delta}_n = \frac{\bar{p}_n - p_0}{p_0} = \frac{\bar{p}_n}{p_0} - 1.0,$$

where $\bar{p}_n$ denotes the average execution price for all q_n shares over $(t_n, t_{n+1}]$:

$$\bar{p}_n = \frac{\sum_{k=1}^{K_n} p_{n,k} \cdot q_{n,k}}{q_n}, \quad \sum_{k=1}^{K_n} q_{n,k} = q_n.$$

Here, $(q_{n,k} \mid 1 \leq k \leq K_n)$ denotes the individual grandchild orders executed by the COP manager over $(t_n, t_{n+1}]$, and $(p_{n,k} \mid 1 \leq k \leq K_n)$ their execution prices.

Let Q denote the size of the entire order, and $Q_{exe} = q_0 + q_1 + \dots + q_{N-1}$ the total executed shares over the designated horizon $(0, T]$. For dynamic trading, generally one could only guarantee that $Q_{exe} \leq Q$. The realized *implementation shortfall* (IS) per dollar is defined to be:

$$IS_{exe} = \frac{\sum_{n=0}^{N-1} \bar{p}_n q_n - p_0 \cdot Q_{exe}}{p_0 \cdot Q_{exe}} = \frac{1}{Q_{exe}} \sum_{n=0}^{N-1} \bar{\delta}_n \cdot q_n \quad (9)$$

For example, $IS_{exe} = 0.0010$ means that to buy every \$10,000 amount of the security (as valued at $t = t_0 = 0$), one actually ends up paying \$10,010 on average.

By Eqn. (9), IS is determined by all the *posterior* moneyiness $\bar{\delta}_n$'s which are unavailable at each planning time t_n . They clearly depend on the observables – the state variables of moneyiness δ_n 's that actually affect trading decisions.

We now introduce the second state variable – the forward PoV ν_n . At the starting time t_n of each arrival bin $(t_n, t_{n+1}]$, we define the residual Q_n via:

$$Q_0 = Q, \quad Q_n = Q_{n-1} - q_{n-1}, \quad 1 \leq n \leq N, \quad (10)$$

where q_{n-1} is the shares acquired over the departing bin $(t_{n-1}, t_n]$. Thus Q_n represents the total shares still needed to be executed at t_n . Let M_n denote the (projected) total *market* volume over the remainder of the execution horizon $(t_n, T]$. In order to minimize complexities, in the current work we assume market volumes obey a deterministic volume “profile”. In many execution houses, such profiles are often automatically generated by overnight or over-weekend processes that utilize the most recent market information. We must point out that for general styled trading, one could also introduce randomness into market volumes and a new state variable associated with it. Such a style can be more realistic but also more complex.

At the starting time t_n of an arrival bin $(t_n, t_{n+1}]$, the forward PoV is defined to be:

$$\nu_n = \frac{Q_n}{M_n}, \quad n = 0, 1, \dots, N-1. \quad (11)$$

Similar to moneyiness δ_n defined earlier, the forward PoV ν_n is also *dimensionless*. Moreover,

$$\nu_0 = \frac{Q_0}{M_0} = \frac{\text{Total Shares to Execute over } [0, T]}{\text{Total Market Shares over } [0, T]}$$

is precisely the PoV for a VWAP algorithm.

4.2. State Transitions and Permanent Market Impacts

Let $V_n = M_n - M_{n+1}$ denote the market volume over $(t_n, t_{n+1}]$. We define the actual PoV over the bin by:

$$u_n = \frac{q_n}{V_n}, \quad \text{which is dimensionless.} \quad (12)$$

For the MV-MVP style being constructed, we define

$$\text{State Variables: } \delta_n, \nu_n; \quad \text{and Control Variables: } u_n. \quad (13)$$

Notice that all three variables are dimensionless.

Let $\rho_n = \frac{V_n}{M_n}$ denote the volume percentage of a “current” bin $(t_n, t_{n+1}]$ over the remaining horizon $(t_n, T]$. The forward PoV has the transition equation:

$$\nu_{n+1} = \frac{Q_{n+1}}{M_{n+1}} = \frac{Q_n - V_n \cdot u_n}{M_n - V_n} = \frac{Q_n/M_n - V_n/M_n \cdot u_n}{1 - V_n/M_n} = \frac{\nu_n - \rho_n \cdot u_n}{1 - \rho_n}, \quad (14)$$

which is linear since ρ_n is known given a volume profile.

For intraday trading, the traditional Brownian motion for dP_t/P_t can be well approximated by the Brownian motion of dP_t/P_0 . Since $d\delta_t = \frac{dP_t}{P_0}$, in the discrete setting one could thus assume that

$$\delta_{n+1} = \delta_n + \sigma_n \sqrt{\Delta t} Z_n + \text{ImpactCost}_n(u_n), \quad (15)$$

where σ_n is the (annualized) spot volatility for $(t_n, t_{n+1}]$, and Z_n an independently drawn canonical Gaussian. Notice that for intraday trading, we always assume that market buys and sells are in balance so that no directional drift exists. The impact cost of the trading of $q_n = u_n \cdot V_n$ shares is $\text{ImpactCost}_n(u_n)$, which affects the execution prices of all trailing trades over $(t_{n+1}, T]$. It is thus called the *permanent impact*. For illustrative purpose, we assume:

$$\text{ImpactCost}_n(u_n) = \mu \cdot \sigma_n \sqrt{\Delta t} \cdot u_n^\gamma, \quad (16)$$

with some bin independent exponent $\gamma \in (0, 1]$ and coefficient $\mu > 0$. It has been broadly noted (e.g., [11]) that permanent costs are concave functions which requires $\gamma \leq 1$. Since δ_n (moneyness), σ_n (annualized spot volatility), and u_n (realized PoV) are all dimensionless, so must be the permanent impact coefficient μ . Our earlier work [17] gives a more detailed account on how to calibrate impact models from trading databases. It is also possible to incorporate the so-called *transient impact* if necessary, though we will not consider it for the MV-MVP style. Transient impact represents the resilient effects of all the preceding trades $u_{n-1}, u_{n-2}, \dots$ by time t_n , and is often expressed as a moving average (see e.g., Obizhara-Wang [12] and Shen [17]). To summarize, we have established the following state transition equations: for bin $n = 0, \dots, N-1$:

$$\nu_{n+1} = \frac{\nu_n - \rho_n \cdot u_n}{1 - \rho_n}, \quad \rho_n = \frac{\nu_n}{M_n} = 1 - \frac{M_{n+1}}{M_n} \quad (17)$$

$$\delta_{n+1} = \delta_n + \sigma_n \sqrt{\Delta t} \cdot Z_n + \mu \sigma_n \sqrt{\Delta t} \cdot u_n^\gamma. \quad (18)$$

The initial state is:

$$\nu_0 = \frac{Q}{M_0}, \text{ the PoV for VWAP,} \quad \text{and} \quad \delta_0 = \frac{p_0 - p_0}{p_0} = 0.0. \quad (19)$$

5. The MV-MVP Style

5.1. The Objective, Instantaneous Impact, and Augmented IS Cost

Let Q_N be the residual shares by the end of an execution horizon $t_N = T$. Generally due to the cap on volume participation imposed by traders, Q_N could be non-zero. The number of executed shares is thus

$$Q_{exe} = Q - Q_N = q_0 + q_1 + \dots + q_{N-1}. \quad (20)$$

By the dimensionless Eqn. (9), the total dollar amount of the IS becomes

$$IS_{exe}^{\$} = p_0 \cdot Q_{exe} \cdot IS_{exe} = p_0 \cdot \sum_{n=0}^{N-1} \bar{\delta}_n \cdot q_n. \quad (21)$$

The *expected* difference between the posterior moneyiness $\bar{\delta}_n$ and the moneyiness state δ_n at t_n is modeled by the *instantaneous* impact. Due to the short span of Δt , we adopt a linear model [17]:

$$\bar{\delta}_n - \delta_n = \eta \cdot \sigma_n \sqrt{\Delta t} \cdot u_n + \eta_{BB} \cdot \sigma_n \sqrt{\Delta t} \cdot Z_n. \quad (22)$$

The details are as follows.

- (1) η represents the instantaneous effect, and is in general inversely proportional to either the liquidity level under macro measures of an security (e.g., average daily volume or spread) or the thickness of the limit order book under market microstructures.
- (2) The canonical Gaussian Z_n is the same as in Eqn. (15), and the η_{BB} -term represents the average random cost of all q_n shares traded over $(t_n, t_{n+1}]$. Thus η_{BB} should be closely correlated to the in-house COP strategies. It can be shown under the theory of Brownian Bridges [10] that if the COP strategies uniformly place individual orders, one has $\eta_{BB} \approx 0.5$.

Due to linearity, η and η_{BB} can be calibrated via linear regression using in-house trading data. To impose the completion condition softly, we define the residual cost:

$$IS_{res}^{\$} = \Gamma \cdot \sigma p_0 \cdot Q_N, \quad (23)$$

where σ denotes the annualized daily volatility, and $\Gamma > 0$ is a residual-averse weight to penalize residuals. For instance, one could set $\Gamma = \Gamma_0 \cdot \sqrt{\frac{1}{BDAYS}}$, where $BDAYS$ denotes the number of trading days per year and Γ_0 an order $O(1)$ constant. This way the residual cost $IS_{res}^{\$}$ is roughly proportional to the overnight holding risk before one could trade the rest on the next business day.

Introduce the execution profile $\mathbf{q} = (q_0, \dots, q_{N-1})$. Combining (21) and (23), we define the total *augmented* IS cost $IS_{\Gamma} = IS_{\Gamma}(\mathbf{q} | Q)$ per dollar to be:

$$IS_{\Gamma} = \frac{1}{Q} \sum_{n=0}^{N-1} \bar{\delta}_n \cdot q_n + \Gamma \cdot \sigma \frac{Q_N}{Q} = \frac{IS_{exe}^{\$} + IS_{res}^{\$}}{Q \cdot p_0} \quad (24)$$

with $Q_N = Q - q_0 - q_1 - \dots - q_{N-1}$. It also depends on the market path $\boldsymbol{\omega} = (Z_0, \dots, Z_{N-1})$, as in Eqn. (15) and (22). Thus one writes $IS_{\Gamma} = IS_{\Gamma}(\mathbf{q}, \boldsymbol{\omega} | Q)$. The objective is to minimize risk adjusted IS in the mean-variance framework of Almgren and Chriss [2, 17]:

$$\min_{\mathbf{q}} E_{\boldsymbol{\omega}}[IS_{\Gamma}(\mathbf{q}, \boldsymbol{\omega} | Q)] + \lambda \cdot \text{VAR}_{\boldsymbol{\omega}}[IS_{\Gamma}(\mathbf{q}, \boldsymbol{\omega} | Q)] \quad (25)$$

for some given level of risk aversion $\lambda > 0$. As discussed earlier, traditional dynamic trading can hardly handle such non-additive objectives.

5.2. Design of The MV-MVP Style

As in Section 4, we shall work with the dimensionless PoV variables $u_n = \frac{q_n}{v_n}$. Define the execution PoV profile $\mathbf{u} = (u_0, \dots, u_{N-1})$. Then the objective expression (25) can also be written as

$$\min_{\mathbf{u}} E_{\omega}[IS_{\Gamma}(\mathbf{u}, \omega \mid Q)] + \lambda \cdot \text{VAR}[IS_{\Gamma}(\mathbf{u}, \omega \mid Q)] \quad (26)$$

The particular model being designed is called the MV-MVP style in order to highlight that (a) it is mean-variance based, and (b) the state variables are given by moneyness and volume participation (via forward PoV). The MV-MVP allows two important features.

- (i) To set a maximum volume participation limit $\text{maxPoV} \in (0, 100\%)$. Most clients prefer a PoV cap like $\text{maxPoV} = 10\%, 15\%$, or 25% .
- (ii) To enforce a *minimum* participation rate minPoV . In DP trading, this is to ensure that at least a minimum amount of shares can be executed over a trading horizon, regardless of the actual market conditions.

Consequently we seek a policy in the form of:

$$u_n = \text{minPoV} \vee h_n \wedge \text{maxPoV} = \min(\max(\text{minPoV}, h_n), \text{maxPoV}), \quad (27)$$

where h_n is the “free” PoV in the linear form of:

$$h_n = \nu_0 + \alpha_n \cdot \delta_n + \beta_n \cdot (\nu_n - \nu_0), \quad n = 0, 1, \dots, N-1. \quad (28)$$

(Be reminded that the running example is for a buy order with $u_n \geq 0$.) Notice that $\nu_0 = \frac{Q_0}{M_0} = \frac{Q}{M_0}$ is exactly the PoV for the VWAP algorithm. Here the two coefficient series (α_n) and (β_n) provide further freedom for style design. To proceed, we always assume this compatibility condition to hold:

$$\text{minPoV} \leq \nu_0 \leq \text{maxPoV}. \quad (29)$$

While maxPoV is typically exposed to clients, minPoV could be internally designed.

We make the following comments regarding the roles of (α_n) and (β_n) :

- (i) The free PoV h_n in Eqn. (28) is like a *factor model* commonly used in risk analytics [14], where δ_n and ν_n are the two observable factors.
- (ii) The coefficients (α_n) effectively express the popular industrial styles of being “aggressive ITM” or “passive ITM”, as discussed in Section 3. For the running example of a buy order, ITM means $\delta_n < 0$. Then a positive α_n would reduce the PoV h_n while being ITM, and thus achieve being “passive ITM”. Similarly, a negative α_n amounts to becoming aggressive ITM.
- (iii) When setting $\alpha_n \equiv 0.0$ and $\beta_n \equiv 1.0$, $n = 0, \dots, N-1$, one has $h_n \equiv \nu_n$. In this scenario one can further show recursively that $u_n \equiv h_n \equiv \nu_n \equiv \nu_0$. That is, the MV-MVP style becomes the canonical VWAP algorithm.

- (iv) Linear policies like Eqn. (28) are actual closed-form solutions in the classical constraint-free LQR models (*Linear Quadrature Regulators*), in which linear state transitions are coupled with stagewise quadratic costs [5].

The computational burden for the policies in Eqn. (27) and (28) under the mean-variance objective (25) is still enormous, as one has to optimize the coefficient series in a high-dimensional space:

$$(\alpha_0, \beta_0; \alpha_1, \beta_1; \dots; \alpha_{N-1}, \beta_{N-1}) \in R^{2N}.$$

More importantly, Bellman-type backward procedures do not apply as the mean-variance objective is not additive.

As a result, we complete the design of the MV-MVP style by further choosing an explicit parametric form for the decision coefficients: for $n = 0, \dots, N-1$,

$$\alpha_n = a \cdot \left[1 - \frac{n}{N-1}\right]^{1+b}, \quad \beta_n = 1 + c \cdot \left[1 - \frac{n}{N-1}\right]^{1-d}, \quad (30)$$

where one only requires $b > -1.0$ and $d < 1.0$, so that as n approaches the last trading bin $N-1$, one has:

$$\alpha_n \rightarrow 0.0, \quad \beta_n \rightarrow 1.0. \quad (31)$$

When $a = c = 0.0$, one recovers the VWAP algorithm.

In general, the MV-MVP style allows more active response to moneyness in the beginning of trading. Near the end of an execution horizon, however, it starts to prioritize in the completion requirement by converging to the forward PoV, i.e., $h_n \rightarrow \nu_n$.

6. Calibration of Styled Trading Models

To summarize, the MV-MVP style eventually becomes an optimization problem with only four unknown parameters (a, b, c, d):

$$\min_{(a,b,c,d)} F(a, b, c, d) = E[IS_\Gamma(\mathbf{u}, \boldsymbol{\omega} \mid Q)] + \lambda \cdot \text{VAR}[IS_\Gamma(\mathbf{u}, \boldsymbol{\omega} \mid Q)]. \quad (32)$$

The policy style is defined via the specific parametric form:

$$\begin{cases} u_n = \min\text{PoV} \vee h_n \wedge \max\text{PoV}, \\ h_n = \nu_0 + \alpha_n \cdot \delta_n + \beta_n \cdot (\nu_n - \nu_0), \\ \alpha_n = a(1 - \frac{n}{N-1})^{1+b}, \quad \beta_n = 1 + c(1 - \frac{n}{N-1})^{1-d}. \end{cases} \quad (33)$$

Also recall that the state transition equations are:

$$\begin{cases} \delta_{n+1} = \delta_n + \mu \cdot \sigma_n \sqrt{\Delta t} \cdot u_n^\gamma + \sigma_n \sqrt{\Delta t} Z_n, \\ \nu_{n+1} = \frac{\nu_n - \rho_n u_n}{1 - \rho_n}, \quad \rho_n = \frac{\nu_n}{M_n} = 1 - \frac{M_{n+1}}{M_n}. \end{cases} \quad (34)$$

The initial state is given by $\delta_0 = 0.0$ and $\nu_0 = \frac{Q}{M_0}$, and the latter is precisely the PoV for the VWAP algorithm.

Finally, as in the preceding section, along any path $\omega = (Z_0, \dots, Z_{N-1})$,

$$\begin{cases} IS_\Gamma(\mathbf{u}, \omega \mid Q) = \frac{1}{Q} \sum_{n=0}^{N-1} \bar{\delta}_n \cdot q_n + \Gamma \cdot \sigma \cdot \frac{Q_N}{Q}, & q_n = u_n \cdot V_n, \\ \bar{\delta}_n = \delta_n + \eta \cdot \sigma_n \sqrt{\Delta t} \cdot u_n + \eta_{BB} \cdot \sigma_n \sqrt{\Delta t} \cdot Z_n. \end{cases} \quad (35)$$

We must remind readers since the model is buy-sell symmetric, we have been consistently working with a buy order for which $Q > 0, u_n, h_n \geq 0$. The setting for a sell order is very similar when working with the absolute or unsigned PoV u_n and absolute order size Q , except for that (i) the plus signs in (34) for the permanent impact term and that in (35) for the instantaneous impact term should both be flipped, and (ii) $\bar{\delta}_n$ in the first term of IS_Γ in (35) should be replaced by $(-\bar{\delta}_n)$. Computationally, one could simply introduce a new variable for order type:

$$side = +1, \text{ for a buy, and } -1, \text{ for a sell,}$$

and place it as a multiplier for the aforementioned three terms.

Let $\Theta = (a, b, c, d)$. Then $IS_\Gamma(\mathbf{u}, \omega \mid Q)$ can be expressed as $IS_\Gamma(\Theta, \omega \mid Q)$, and the objective (32) becomes:

$$\min_{\Theta} F(\Theta) = E[IS_\Gamma(\Theta, \omega \mid Q)] + \lambda \cdot \text{VAR}[IS_\Gamma(\Theta, \omega \mid Q)]. \quad (36)$$

Eventually a styled dynamic trading model becomes a parametric model. Due to the nonlinearities associated with the flooring or capping operators, the power exponents in Eqn. (33), as well as the nonlinearities in the state equations (34), the objective function $F(\Theta) = F(a, b, c, d)$ is generally nonlinear and non-convex. Thus only local optima are sought after via suitably designed iterative schemes. There exists a wealth of literature on optimizing objectives of stochastic nature, and below we briefly describe two major numerical schemes that we have experimented with: *stochastic gradient descent* and *binomial-tree* based gradient descent.

For any given set of styled parameters Θ , the objective involves two ensemble statistics - the mean and the variance. Thus one could employ the method of *stochastic gradient descent* (SGD) by building gradient descending schemes upon Monte-Carlo valuation [16]. Our extensive experiments have shown that the local optima captured this way are sensitive to initial guesses, and that the convergence usually takes long. Conventional variance reduction techniques such as the antithetic method do *not* work well since the objective is *not* monotonic against random market moves.

Alternatively, the binomial-tree based gradient descent (BT-GD) is more stable and works particularly well for horizons with a moderate number of bins, e.g., 90 minutes in the form of 18 bins of 5-minute child windows. In this implementation, the Gaussian market path in Eqn. (35)

$$\omega = (Z_0, \dots, Z_{N-1})$$

is approximately implemented by the binomial path:

$$\omega = (U_0, \dots, U_{N-1}), \quad U_k \sim B(\{1, -1\}, 0.5), \quad k = 0, \dots, N-1,$$

where U_k 's are i.i.d.'s of a symmetric Bernoulli type $B(\{1, -1\}, 0.5)$:

$$\text{Prob}(U_k = 1) = \text{Prob}(U_k = -1) = 0.5.$$

All binomial paths are equally likely under this setting, and the objective function for any given parameter set Θ can be evaluated exactly.

At each iteration step with the “current” estimator $\Theta^0 = (\theta_1^0, \theta_2^0, \theta_3^0, \theta_4^0)$ (corresponding to (a, b, c, d)), the gradient is estimated via *synchronized* finite differencing as follows. Define the basis directions:

$$\mathbf{e}_1 = (1, 0, 0, 0), \quad \dots, \quad \mathbf{e}_4 = (0, 0, 0, 1).$$

For a set of small parameters $(\varepsilon_1, \dots, \varepsilon_4)$ fixed throughout the iteration, the gradient components are estimated via the central difference scheme:

$$\frac{\partial F(\Theta^0)}{\partial \theta_i} = [F(\Theta^0 + \varepsilon_i \mathbf{e}_i) - F(\Theta^0 - \varepsilon_i \mathbf{e}_i)] / 2\varepsilon_i, \quad i = 1, 2, 3, 4.$$

The evaluation of the eight objective values $F(\Theta^0 \pm \varepsilon_i \mathbf{e}_i)$, $i = 1, 2, 3, 4$ is synchronized over all market paths. That is, for each evolving market path ω , the eight IS_Γ 's associated with the eight sets of trading parameters are simultaneously computed and recorded. This improves computational efficiency.

As mentioned earlier, the VWAP strategy is a special instance of the MV-MVP style when:

$$a = b = 0 \text{ (i.e., } \alpha_n \equiv 0), \quad \text{and} \quad c = d = 0 \text{ (i.e., } \beta_n \equiv 1).$$

This provides a natural set of initial guess for any iterative scheme.

Below we demonstrate the behavior of the proposed MV-MVP model via one generic numerical instance. We explain it via a pseudo *object-oriented* fashion, in order to facilitate popular production languages such as C++, Java, or Python.

(A) We assume that the `market` object has the following data fields:

`market.dayInMinutes=390, market.yearInBusinessDays=252.`

This is to simulate a *regular* US equities market which trades from 9:30 am to 4:00 pm (except for special half-days), and has about 252 trading days per year.

(B) For illustrative purposes, we assume *flat* profiles of intraday volatilities and volumes, and the particular `security` object contains this information:

`security.ID="ABC", security.vol=25.0%, security.ADV=4,000,000.`

Here the *average daily volume* (ADV) of 4 million shares represents roughly the average ADV of S&P 500 stocks, which are large-cap and popularly traded. As common in finance, the volatility is annualized. An annual volatility of 25.0% is roughly the average level of the S&P 500 universe.

- (C) The trading object `trade` summarizes both the external trading request from a client and internal trading conventions. For the particular instance, it is assumed that:

```
trade.startIdx=30, trade.endIdx=120, trade.binWindow=5,
trade.maxPoV=25%, trade.minPoV=2.5%,
trade.tradeSize=92300, trade.side="BUY".
```

That is, the trade request is to buy 92300 shares from 10:00am to 11:30am, with PoV capped at 25.0% and floored at 2.5%. The algorithm also internally sets bin size to 5 minutes. The order size corresponds to the 10% PoV for a VWAP strategy over the same horizon.

- (D) The model object `model` is assumed to have the following data fields. In reality, these impact model parameters have to be calibrated from the real trading database. Our earlier work [17] discusses the dependency of these parameters on the *child order placement* (COP) strategies of individual houses. In reality, risk aversion λ is internally mapped to the user-friendly external GUI fields like “Urgency Level.”

`model. $\mu$  = 0.5, model. $\eta$  = 1.0, model. $\Gamma$  = 10.0, model. $\lambda$  = 0.01.`

We have set $\eta_{BB} = 0.5$ as explained earlier.

Given the above instance of the `market`, `security`, `trade`, and `model`, our binomial-tree based calibrator converges at the following set of policy parameters:

$a = 0.23$, $b = -0.48$; $c = 2.24$, $d = 0.48$.

The resulting policy coefficients of α_n ’s and β_n ’s are plotted in Figure 1 via the Eqn. (30).

To better visualize the calibrated MV-MVP style, in Figure 2 we have plotted the simulated trading paths given 10 or 100 independently simulated market paths, according to the dynamic policies in Eqn. (27) and (28). As expected, the MV-MVP style oscillates around the VWAP strategy.

The above computational schemes have resulted from our initial investigations. By no means they are claimed to have achieved the desired efficiency. We thus encourage both academic researchers and industrial practitioners to help mature the computation.

7. Disclaimers

- (1) The proposed trading model has not been the internal or external product of any execution houses where the author had worked. Any potential industrial conflict should be promptly directed to the attention of the author.
- (2) Due to the proprietary nature of the industry and the resulting scarcity of real trading data to the public, the current academic work does not dive deeper into the data aspects. Interested execution houses are thus encouraged to further tailor the current framework to the available trade or market databases, e.g., to use dynamically projected volumes.

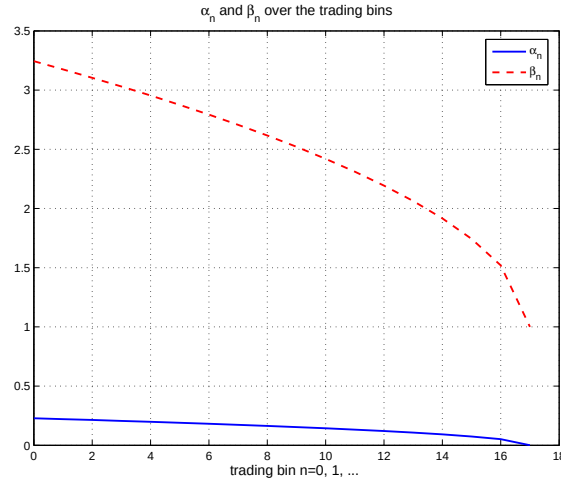


Figure 1: The policy coefficients α_n 's and β_n 's from the calibrated parameters a, b, c , and d .

- (3) Any mentioning of certain brand or company names, e.g., the CPLEX Optimizer, or MOSEK, etc., is not a product endorsement from the author for purchasing or investment, but an indication of some popular practices in the contemporary industry.
- (4) Execution houses who are interested in the current work and plan to implement it in their systems should be aware of any other securities or operational risks, including for examples, IPO's (initial public offerings) without much trading histories, complex market opens or closes, trading halts, or any extreme market events, etc.

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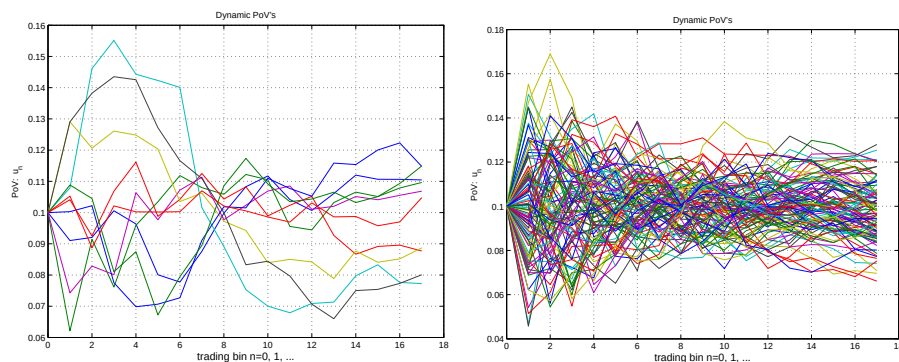


Figure 2: Simulated trading paths via the policies proposed in Eqn. (33) for the calibrated MV-MVP style. Left: 10 trading paths; Right: 100 trading paths. Both are based on independently simulated market paths. As a reference, pay attention that the original total position has been designed so that the VWAP PoV is 10%. The simulated trading paths oscillate around VWAP for different market scenarios as expected. The MV-MVP incorporates a new degree of dynamic response to the moneyness.

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